

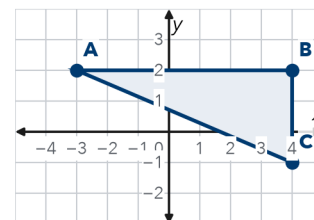
LESSON 7-7 • SESSION 1 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

Plot and Connect Points

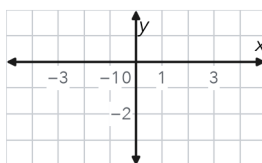
Builds on Lesson 7-5 Session 1 • 6.NS.C.6c

TODAY'S LESSONYou will plot garden corners like $(-4, 6)$ and $(9, -3)$, then connect them.**YOU NEED TO**Plot a point like $(-4, 6)$: move left 4, then up 6. **You will use it in Practice Form A, problems 2 and 4.****STEPS TO FOLLOW***use these same steps on every problem*

- 1** Start at $(0, 0)$. Move **left or right** the first number.
- 2** Move **up or down** the second number. Draw a dot.
- 3** **Connect** the dots in order, then back to the first.

WATCH ME*the steps in action***Plot $A(-3, 2)$, $B(4, 2)$ and $C(4, -1)$. Connect them.****Step 1** A: left 3. B: right 4. C: right 4.**Step 2** A: up 2. B: up 2. C: down 1. Draw dots.**Step 3** Connect A to B to C, then back to A.**Answer** A triangle.**TRY IT WITH HELP***fill in each step*

- 1.** Plot $P(-4, 1)$, $Q(2, 1)$ and $R(2, -3)$. Connect them.

**Step 1** P: _____ 4. Q: right 2. R: right _____.**Step 2** P: up 1. Q: up _____. R: _____ 3.**Step 3** Connect P to Q to _____, then back to _____.

- 2.** Plot $J(-1, 4)$, $K(3, 4)$, $L(3, -2)$ and $M(-1, -2)$. Connect them.

Step 1 J and M: _____ 1. K and L: right _____.**Step 2** J and K: up _____. L and M: _____ 2.**Step 3** Connect J, K, L, M, then back to _____. The shape is a _____.**YOUR TURN***use the steps on your own*

- 3.** Write the two moves from $(0, 0)$ for each point.

a. $(-5, 2)$ _____**b.** $(3, -6)$ _____**c.** $(-1, -4)$ _____**d.** $(0, 7)$ _____

Find Distance on a Grid

Builds on Lesson 7-6 Session 1 • 6.NS.C.8

TODAY'S LESSONIn problem 1, the side from $(-3, 4)$ to $(6, 4)$ is $3 + 6 = 9$ units long.**YOU NEED TO**Find the distance between two points that share a coordinate. **You will use it in Practice Form A, problems 1, 2 and 3.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Find** the shared coordinate. Use the other two numbers.
- 2 Signs** differ? Add the distances from 0. Same sign? Subtract.
- 3 Write** the distance in units.

WATCH ME*the steps in action***Find the distance from $(-4, 3)$ to $(5, 3)$.****Step 1** They share $y = 3$. Use -4 and 5 .**Step 2** Signs differ: $4 + 5 = 9$.**Step 3** The distance is 9 units.**Answer 9 units****TRY IT WITH HELP***fill in each step***4.** Find the distance from $(-6, 1)$ to $(2, 1)$.**Step 1** They share _____ = 1. Use -6 and 2 .**Step 2** Signs differ: $6 + 2 =$ _____.**Step 3** The distance is _____ units.**5.** Find the distance from $(-7, 2)$ to $(-1, 2)$.**Step 1** They share $y = 2$. Use _____ and _____.**Step 2** Same sign: $7 - 1 =$ _____.**Step 3** The distance is _____ units.**YOUR TURN***use the steps on your own***6.** Find the distance between the points in units.**a.** $(-3, 4)$ to $(6, 4)$ ____**b.** $(6, 4)$ to $(6, -2)$ ____**c.** $(2, 8)$ to $(2, 3)$ ____**d.** $(-5, -4)$ to $(-5, 3)$ ____**e.** $(-9, 1)$ to $(-2, 1)$ ____**f.** $(-4, -3)$ to $(9, -3)$ ____**READY CHECK***a problem like the one in today's lesson***7.** A rectangle's top side joins $(-9, 5)$ and $(6, 5)$. How long is the top side?**A** 3 units**B** 15 units**C** 10 units**D** 11 units**CHECK YOURSELF**
☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____